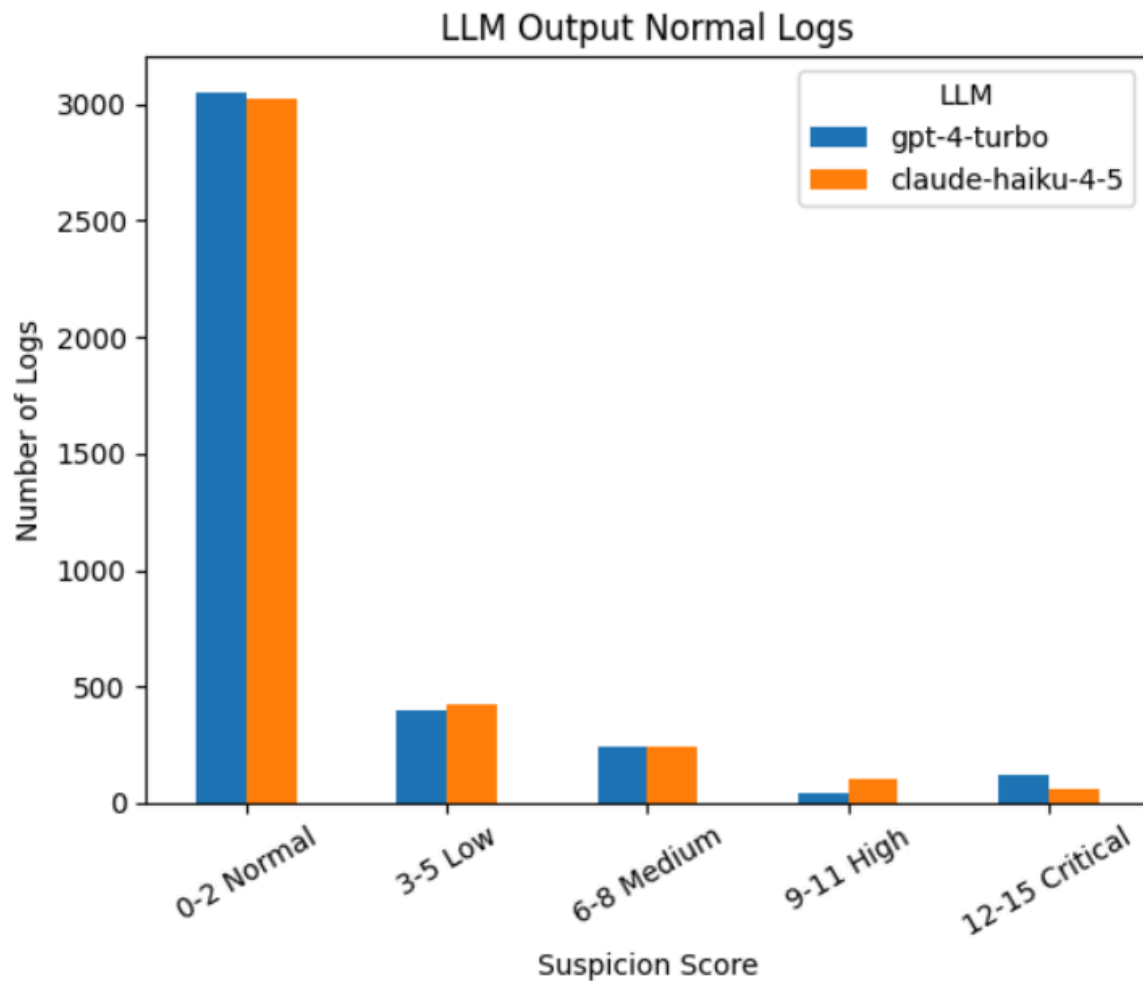
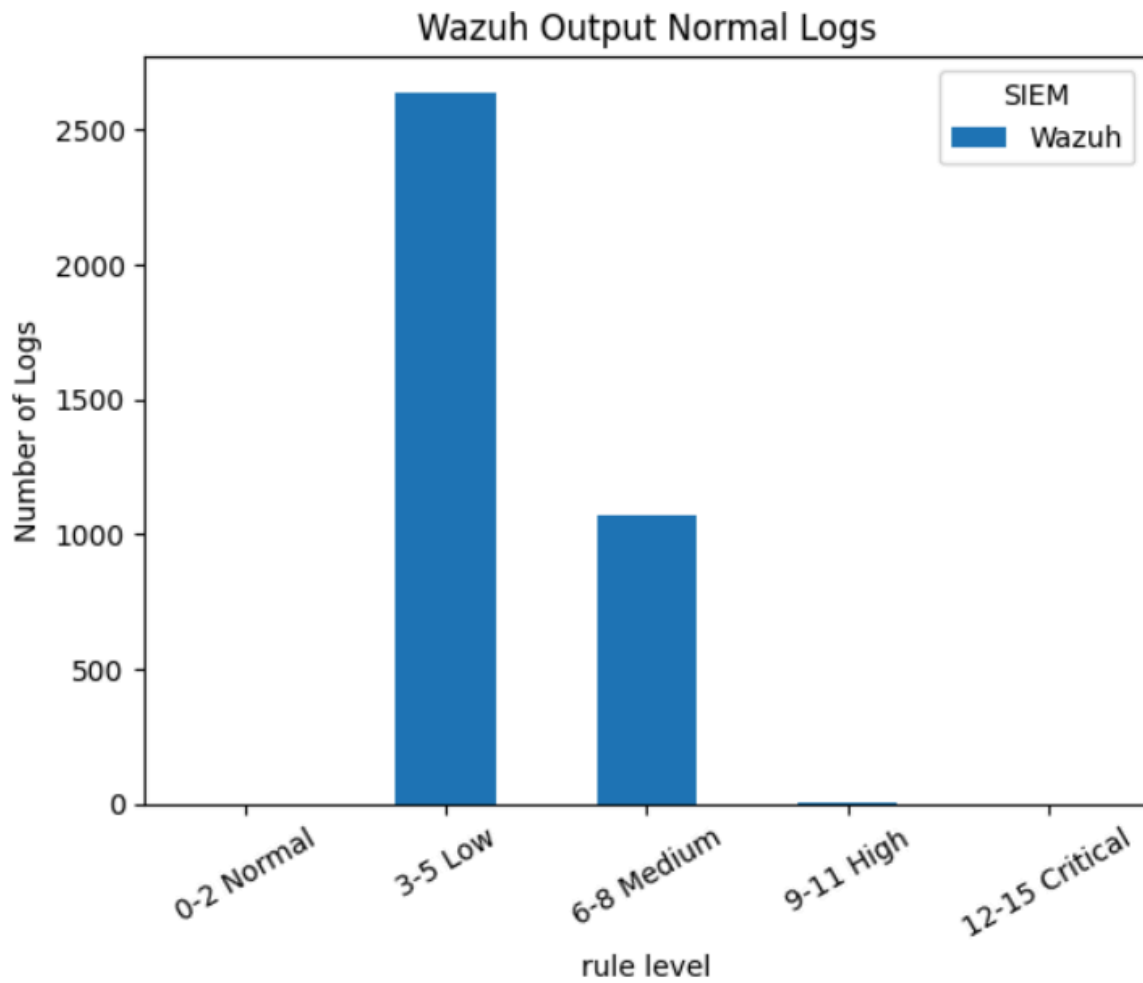


Analysis

Wazuh performs differently from the LLMs used in this project, so it is used as a general baseline for comparison. Wazuh uses a rule-based scoring system. The LLMs were asked to review logs and rate how suspicious they were on a scale from 0 to 15. The LLMs were given the same scale as Wazuh.



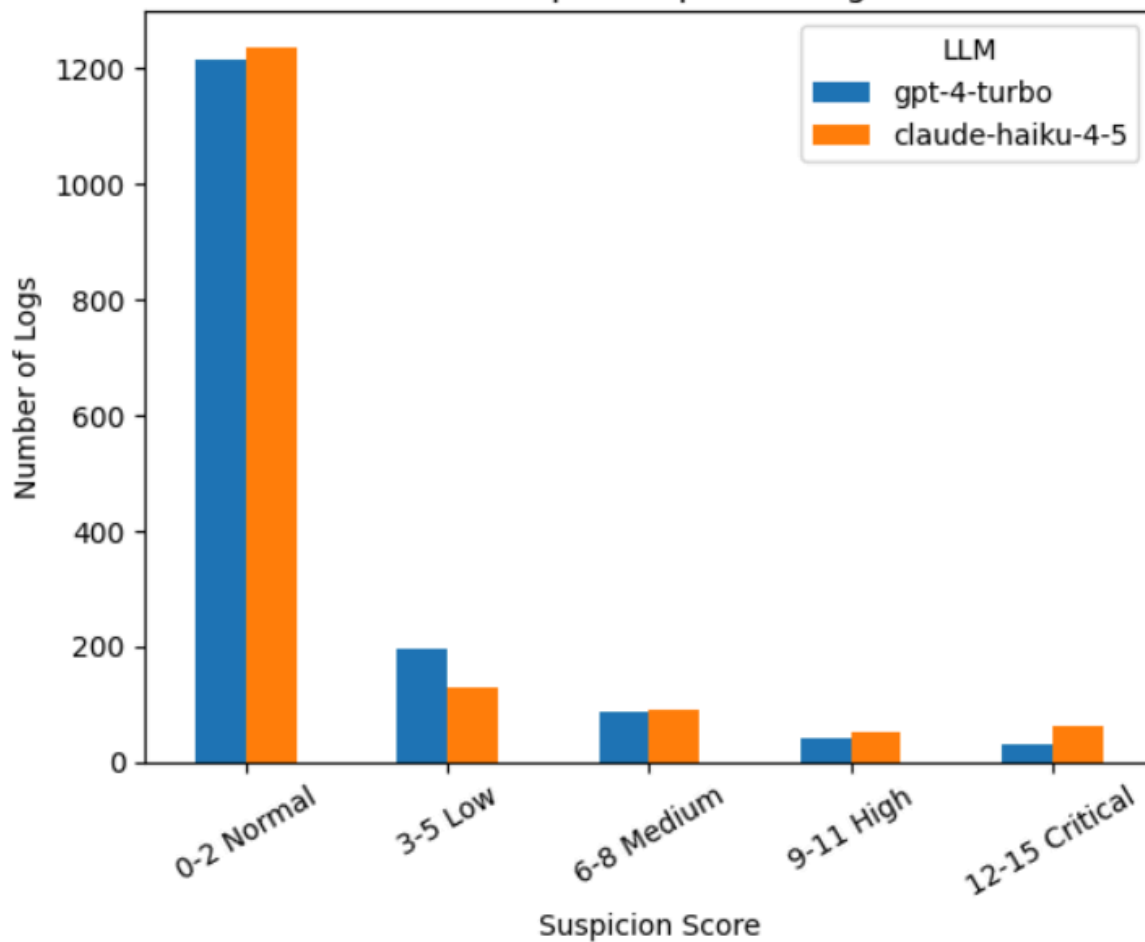
Overall, both LLMs showed fairly similar distributions in how they scored the logs. However, both models also tended to score normal logs as more suspicious than they were. Sometimes, assigning them a higher suspicion score even when there was nothing suspicious in the logs.



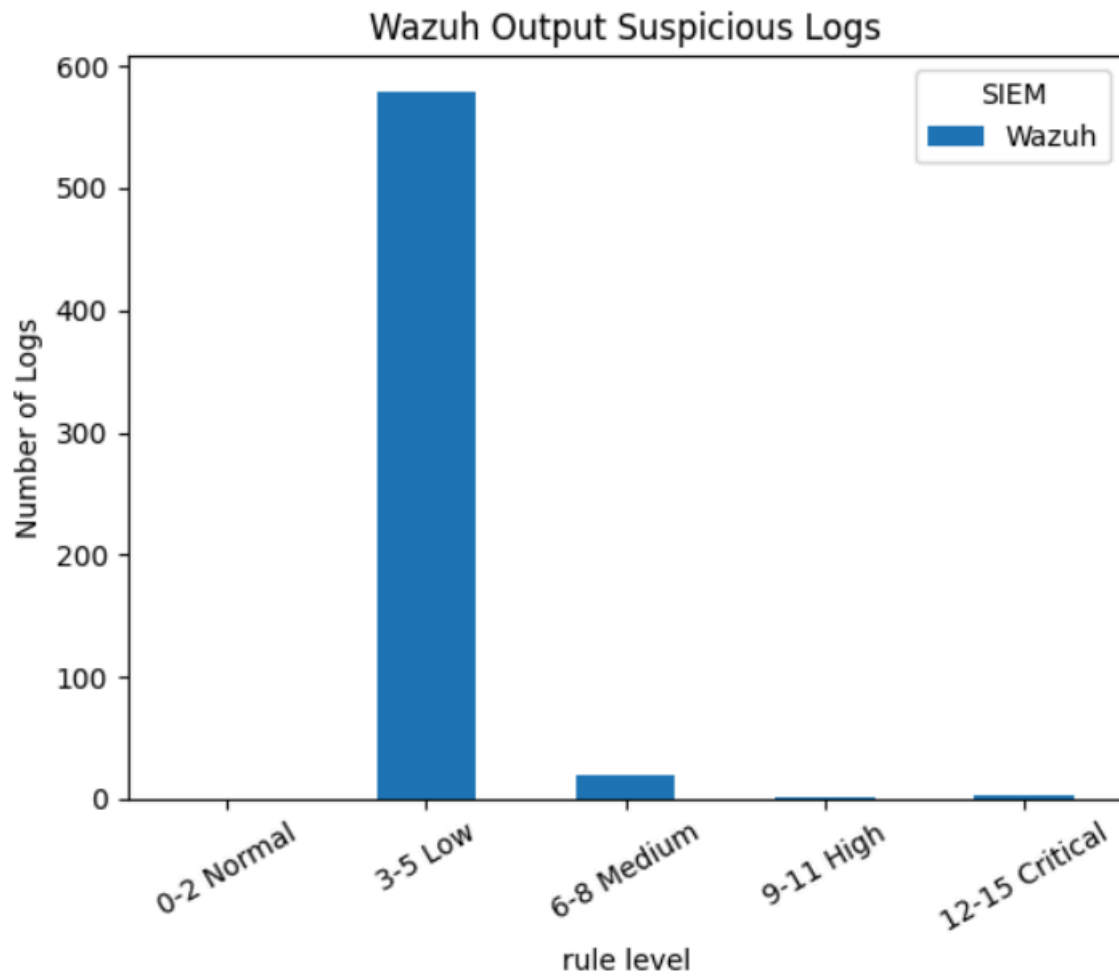
Wazuh had no high and critical alerts for the logs. Wazuh had mostly medium and low alerts and no 0-2 alerts because Wazuh does not create alerts for those logs.

Suspicious Activity

LLM Output Suspicious Logs



Overall, both LLMs showed fairly similar distributions in how they scored the logs. However, both models again tended to score normal logs as more suspicious than they were. Sometimes, assigning them a higher suspicion score even when there was nothing suspicious in the log. The suspicious dataset does have higher and more critical log scores compared to the normal dataset.



Wazuh mostly had low and medium rule level alert. However, for the suspicious dataset, there were a few high and critical rule-level alerts.

Top 5 Results

Normal Dataset

Claude

RowID	Suspicion_	summary
194	13	svchost using suspicious email credential pizzahuthacker@gmail.com
663	11	MSI installer from PowerShell temp directory
742	13	wazuh-agent spawning powershell querying password policy
83	7	Upfc.exe with suspicious launchtype boot parameter
679	8	ICACLS reset permissions ossec-agent directory

GPT

RowID	Suspicion_	summary
194	15	Suspicious email in username
597	15	User added to Administrators
742	10	Security policy export and deletion
708	9	Critical file permission removal
83	12	Unusual executable path

Wazuh

rule\level	rule\id	rule\description
11	521	Possible kernel level rootkit
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)
9	19011	CIS Benchmark for Amazon Linux 2023 Benchmark v1.0.0.: Ensure message of the day is configured
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)

For GPT, all events that were in the top 5 did match the logs. However, the suspicious score for the logs was much higher than it should have been. The summaries for the logs also match closely with what was expected.

For Claude, there were several hallucinations of events that had not occurred. There was no user added to administrators, and no critical file permissions were removed. The summaries from Claude were noticeably less detailed compared to GPT.

Wazuh had a very low level alert for these logs. The VMs were unhardened, and there were several benchmark warnings for being out of compliance. There was a possible kernel-level rootkit warning, but this is due to the VMs not being hardened.

Overall, for the normal dataset, the LLMs were saying normal activity was very suspicious and hallucinating events that never happened. One other thing of note was that the email for the Windows account was pizzahuthacker, and that was being flagged

as a highly suspicious event. The account, however, only did normal activity. The LLMs were overemphasizing the name of the account and not looking at the context of what it was doing. For the top 5, the prompt was changed to have less emphasis on the name of emails; otherwise, the entire top 5 would have been the email's username.

Suspicious dataset

Claude

RowID	Suspicion_	summary				
1371	15	Adding global group to administrators via net				
1372	15	Net1 adding global to administrators group				
1377	15	Adding system2 user to administrators group				
1367	14	Scheduled task creation for persistence via cmd				
1447	14	PowerShell creating scheduled task Updater				

GPT

RowID	Suspicion_	summary				
718	12	Suspicious user login attempt				
1367	12	Suspicious scheduled task creation				
1227	12	Suspicious executable in SoftwareDistribution				
318	12	Suspicious PowerShell command				
1361	12	Suspicious conhost forced execution				

Wazuh

rule\level	rule\id	rule\description				
12	60154	Administrators Group Changed				
12	60154	Administrators Group Changed				
12	60154	Administrators Group Changed				
10	60204	Multiple Windows Logon Failures				
9	19011	CIS Benchmark for Amazon Linux 2023 Benchmark v1.0.0				

For Claude, it successfully detected the suspicious activity and did not have any hallucinations in the top 5. Additionally, the summaries did match closely with what was done in the attack.

For GPT, it successfully detected the suspicious activity, but it did have hallucinations in the top 5. There was no suspicious executable installed on the host. The summaries did match the suspicious activity in the logs.

Wazuh had several alerts for changes in the administrators' group and multiple Windows login failures. Wazuh detected the suspicious activity as expected.

Overall, both LLMs detected suspicious activity and created useful summaries for why it was suspicious. Claude did say the suspicious logs were much more suspicious compared to GPT.

Estimated cost to use LLM for detecting suspicious activity, creating summaries of why logs are suspicious, and alerting for the top 5 most suspicious logs.

The normal dataset was 3,860 rows.

The suspicious dataset had 1,571 rows.

The dataset was collected over a period of 20 minutes.

GPT-4-turbo costs \$8.77 to analyze the normal dataset.

It is estimated that it will cost \$631.44 per day per host.

It is estimated that it will cost \$230,475.60 per year per host.

GPT-4-turbo costs \$4.19 to analyze the suspicious dataset.

It is estimated that it will cost \$301.68 per day per host.

It is estimated that it will cost \$110,113.2 per year per host.

Claude-haiku-4-5 costs \$3.37 to analyze the normal dataset.

It is estimated that it will cost \$242.64 per day per host.

It is estimated that it will cost \$88,563.60 per year per host.

Claude-haiku-4-5 costs \$0.82 to analyze the suspicious dataset.

It is estimated that it will cost \$59.04 per day per host.

It is estimated that it will cost \$21,549.90 per year per host.

Estimating the cost to use LLM for this purpose would be millions to hundreds of millions of dollars per year for companies of all sizes. For this project, I was analyzing how well LLMs detect suspicious activity, creating summaries of why logs are suspicious, and alerting for the top 5 most suspicious logs. I was sending close to all the log data to the LLM to understand and evaluate performance. In a company, this would not be done. Normally, LLMs are paired with a SIEM, and only suspicious activity would be sent to the LLM, which would greatly reduce the cost. This was not done for this project due to the time constraints and complexity of implementing the system. Also, generally, an LLM would not be used to find suspicious activity, but to write reports on what happened. A SIEM that uses LLMs is Microsoft Sentinel. The pricing is per GB. The pricing varies widely depending on the size of the company. For each 1 GB, it is \$4.30. Companies can also buy in bulk for a discounted price. For the top tier, they can get 50,000 GB for \$102,600 per day. Overall, this project shows that LLMs can help make logs easier to understand and reduce alert fatigue, but they work best when used alongside traditional detection systems instead of on their own.

<https://www.microsoft.com/en-us/security/pricing/microsoft-sentinel>